A STRETCHED-EXPONENTIAL LOWER BOUND FOR ERDŐS PROBLEM 539

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Abstract. For a finite set A of positive integers, let

$$Q(A) = \left\{ \frac{a}{\gcd(a,b)} : a, b \in A \right\}, \qquad h(n) = \min_{|A|=n} |Q(A)|.$$

We prove that there is an absolute constant $c > 0$ such that

$$h(n) \geq \sqrt{n}\, \exp\bigl(c(\log n)^{1/3}\bigr)$$

for all sufficiently large n. Via prime-valuation vectors, the problem is equivalent to bounding

$$D(F) = \{(x-y)^+ : x, y \in F\}$$

for finite lattice sets F. The main new ingredient is a rooted bipartite codegree-amplification lemma. It yields the fixed-dimensional estimate

$$|D(F)| \geq e^{-4d} |F|^{1/2+1/(4d^2-2)} \qquad (F \subset \mathbb{R}^d).$$

A weak Polynomial Freiman–Ruzsa theorem then reduces any putative near-extremizer to affine dimension $O(\log(|D(F)|/\sqrt{|F|}))$, and the two estimates combine to give the stated stretched-exponential lower bound.

1. Introduction

For a finite set $A \subset \mathbb{Z}_{>0}$, define

$$Q(A) = \left\{ \frac{a}{\gcd(a,b)} : a, b \in A \right\}, \qquad h(n) = \min_{|A|=n} |Q(A)|.$$

Erdős Problem 539 asks for the asymptotic size of $h(n)$. The currently documented dimension-free estimates are

$$\frac{1+\sqrt{8n-7}}{2} \leq h(n) \leq \sqrt{n}\, \exp\bigl(C\sqrt{\log n}\bigr), \tag{1}$$

so in particular $h(n) = n^{1/2+o(1)}$; see [3, Appendix A]. The polynomial exponent is therefore known, but the subpolynomial factor multiplying $\sqrt{n}$ remains unresolved.

Our main result strengthens the lower bound by an unbounded stretched-exponential factor.

Theorem 1.1. *There is an absolute constant $c > 0$ such that*

$$\boxed{h(n) \geq \sqrt{n}\, \exp\bigl(c(\log n)^{1/3}\bigr)}$$

for all sufficiently large n.

The theorem is a lower bound in the literal extremal sense: the proof begins with an arbitrary n-element set A, proves the asserted inequality for that set, and only at the end takes the minimum over all A.

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This manuscript is an unrefereed preprint prepared for independent mathematical verification. The proof is non-computational; supplementary scripts check algebraic bookkeeping and selected finite instances only.

Outline of the proof. The argument has four parts.

(1) Prime valuations convert the arithmetic expression $a/\gcd(a,b)$ into a positive-part difference $(x-y)^+$ of lattice vectors.
(2) A mixed coordinate orthant is represented as a rooted bipartite graph. Common neighborhoods of right-side vertices create disjoint "shelves" of positive differences.
(3) The shelf argument gives, in every fixed dimension d,
$$|D(F)| \geq e^{-4d}|F|^{1/2+1/(4d^2-2)}.$$
(4) If $|D(F)| = L\sqrt{|F|}$, then F has ordinary additive doubling at most L^6. Weak Polynomial Freiman–Ruzsa theory supplies a polynomially large subset of affine dimension $O(\log L)$. Applying the fixed-dimensional bound in that dimension forces
$$\log L \gg (\log |F|)^{1/3}.$$

All logarithms are natural. Exact finite inequalities are kept distinct from asymptotic notation throughout.

2. The lattice formulation

For $z = (z_1, \ldots, z_d) \in \mathbb{R}^d$, write
$$z^+ = (\max(z_1, 0), \ldots, \max(z_d, 0)).$$
For a finite set $F \subset \mathbb{R}^d$, define
$$D(F) = \{(x-y)^+ : x, y \in F\}.$$

Proposition 2.1 (Prime-valuation reformulation)**.** *For every $n \geq 1$,*
$$h(n) = \min_{\substack{d \geq 1,\ F \subset \mathbb{Z}^d \\ |F|=n}} |D(F)|.$$

Proof. Let $p_1, \ldots, p_d$ be the primes occurring in a finite set A and write
$$a = \prod_{i=1}^{d} p_i^{x_i}, \qquad b = \prod_{i=1}^{d} p_i^{y_i}.$$
Then
$$v_{p_i}\left(\frac{a}{\gcd(a,b)}\right) = x_i - \min(x_i, y_i) = \max(x_i - y_i, 0).$$
Thus the valuation vector of $a/\gcd(a,b)$ is exactly $(x-y)^+$, and unique factorization gives
$$|Q(A)| = |D(F)|$$
for the corresponding exponent-vector set F.

Conversely, let $F \subset \mathbb{Z}^d$ be finite. Choose $t \in \mathbb{Z}^d$ so that $F + t \subset \mathbb{Z}^d_{\geq 0}$. Translation does not change positive differences:
$$((x+t) - (y+t))^+ = (x-y)^+.$$
Encoding $x+t$ by $\prod_{i=1}^d p_i^{x_i+t_i}$ with distinct primes p_i produces a set of positive integers having exactly the same positive-difference set. This proves the identity. □

We record three elementary facts.

Lemma 2.2. *Let $F \subset \mathbb{R}^d$ be finite and nonempty.*

(i) $F - F \subset D(F) - D(F)$.
(ii) $|D(F)|^2 \geq 2|F| - 1$, *and hence* $|D(F)| \geq |F|^{1/2}$.

(iii) *If π deletes coordinates, then*
$$\pi((x-y)^+) = (\pi x - \pi y)^+,$$
so $D(\pi F) = \pi(D(F))$ and $|D(\pi F)| \le |D(F)|$.

Proof. Part (i) follows from
$$x - y = (x-y)^+ - (y-x)^+.$$
For (ii), choose a linear functional that is injective on F and order the points as $x_1, \dots, x_m$ by increasing functional value. The m differences $x_i - x_1$ and the $m-1$ differences $x_1 - x_i$ for $i \ge 2$ are distinct, so $|F - F| \ge 2m - 1$. Now apply (i). Part (iii) is coordinatewise. □

3. Rooted orthant graphs

Let $X \subset \mathbb{R}^p_{\ge 0}$ and $Y \subset \mathbb{R}^r_{\ge 0}$ be finite active vertex sets, and let $G \subset X \times Y$ be a bipartite graph. Regard an edge (u, v) as the point $(u, -v) \in \mathbb{R}^{p+r}$ and define
$$\Delta(G) = \left\{ \left((u-u')^+, (v'-v)^+\right) : (u,v), (u',v') \in G \right\}. \tag{2}$$
If $(0,0) \in G$, we say that G is rooted.

Lemma 3.1 (Root-axis bounds)**.** *If G is rooted and $x = |X|$, $y = |Y|$, then*
$$|\Delta(G)| \ge x, \qquad |\Delta(G)| \ge y.$$

Proof. Comparing (u, v) with the root in the two orders gives
$$(u, 0) \in \Delta(G), \qquad (0, v) \in \Delta(G).$$
Active supports make these x and y distinct outputs, respectively. □

4. Rooted codegree amplification

The following lemma is the main combinatorial input.

Lemma 4.1 (Rooted codegree amplification)**.** *Let $1/2 < \alpha \le 1$ and $0 < c \le 1$. Suppose every finite nonempty $W \subset \mathbb{R}^p$ satisfies*
$$|D(W)| \ge c|W|^\alpha. \tag{3}$$
Let $G \subset X \times Y$ be rooted, with $X \subset \mathbb{R}^p_{\ge 0}$, $Y \subset \mathbb{R}^r_{\ge 0}$, and write
$$x = |X|, \qquad y = |Y|, \qquad |G| = \delta x y.$$
Then
$$\boxed{|\Delta(G)| \ge \frac{c}{16}\sqrt{2\alpha - 1}\, \delta^{2\alpha} x^\alpha y^{1/2}.} \tag{4}$$

Proof. Put
$$q = |\Delta(G)|, \qquad \beta = 2\alpha - 1 \in (0, 1].$$
By Lemma 3.1, $q \ge x$ and $q \ge y$.

We first dispose of two sparse cases. If $\delta y < 2$, then, since $x^\alpha \le x$, $y^{1/2 - 2\alpha} \le 1$, and $2^{2\alpha} \le 4$,
$$\frac{c}{16}\sqrt{\beta}\, \delta^{2\alpha} x^\alpha y^{1/2} = \frac{c}{16}\sqrt{\beta}\, x^\alpha (\delta y)^{2\alpha} y^{1/2 - 2\alpha} \le \frac{x}{4} \le q.$$
If $\delta^2 x < 1$, then
$$\frac{c}{16}\sqrt{\beta}\, \delta^{2\alpha} x^\alpha y^{1/2} = \frac{c}{16}\sqrt{\beta}\, (\delta^2 x)^\alpha y^{1/2} \le \frac{y^{1/2}}{16} \le q.$$
Hence assume
$$\delta y \ge 2, \qquad \delta^2 x \ge 1. \tag{5}$$

For $v, v' \in Y$, let
$$\kappa(v, v') = |N(v) \cap N(v')|.$$

Double counting and Cauchy–Schwarz give

$$\sum_{v,v'\in Y} \kappa(v,v') = \sum_{u\in X} \deg(u)^2 \geq \frac{|G|^2}{x} = \delta^2 xy^2. \tag{6}$$

The diagonal contribution is

$$\sum_{v\in Y} \kappa(v,v) = |G| = \delta xy.$$

Let K be the unordered off-diagonal codegree sum. From (5) and (6),

$$K = \frac{1}{2}\left(\sum_{v,v'} \kappa(v,v') - |G|\right) \geq \frac{1}{4}\delta^2 xy^2. \tag{7}$$

Set

$$t = K/y^2.$$

For $j \geq 0$, put

$$s_j = 2^j t/4$$

and let E_j be the set of unordered pairs $\{v,v'\}$ satisfying

$$s_j \leq \kappa(v,v') < 2s_j.$$

Write $e_j = |E_j|$. The pairs with $\kappa(v,v') < t/4$ contribute at most

$$\binom{y}{2}\frac{t}{4} \leq \frac{y^2 t}{8} = \frac{K}{8}$$

to K. Consequently the pairs assigned to the levels E_j carry codegree mass at least $7K/8$. Since $\kappa(v,v') < 2s_j$ on E_j,

$$\sum_{j\geq 0} s_j e_j \geq \frac{7K}{16}. \tag{8}$$

For each j, define

$$C_j = \left\{(v-v')^+, (v'-v)^+ : \{v,v'\} \in E_j\right\}, \qquad R_j = |C_j|.$$

There are $2e_j$ oriented pairs associated with E_j. An oriented pair determines the ordered pair of positive parts

$$\big((v'-v)^+, (v-v')^+\big),$$

which in turn determines the exact difference $v'-v$. A fixed exact difference has at most y ordered representations in Y. Hence

$$2e_j \leq yR_j^2. \tag{9}$$

Fix $a \in C_j$ and choose an oriented pair (v,v') with

$$a = (v'-v)^+.$$

Its common neighborhood $W = N(v) \cap N(v')$ has at least s_j elements. For $u,u' \in W$, the edges (u,v) and (u',v') produce

$$\big((u-u')^+, a\big) \in \Delta(G).$$

Thus $D(W) \times \{a\}$ is a shelf in $\Delta(G)$ of size at least cs_j^α. Shelves with distinct second coordinate a are disjoint. Therefore

$$q \geq cR_j s_j^\alpha. \tag{10}$$

Combining (9) and (10),

$$e_j \leq \frac{yq^2}{2c^2 s_j^{2\alpha}}. \tag{11}$$

Using (8) and (11),

$$\begin{aligned}\frac{7K}{16} &\le \sum_{j\ge 0} s_j e_j \\ &\le \frac{yq^2}{2c^2}\sum_{j\ge 0} s_j^{1-2\alpha} \\ &= \frac{yq^2}{2c^2}\left(\frac{t}{4}\right)^{-\beta}\frac{1}{1-2^{-\beta}}.\end{aligned}$$

For $0<\beta\le 1$, the elementary inequality $1-e^{-z}\ge z/2$ for $0\le z\le 1$, applied with $z=\beta\log 2$, gives

$$1-2^{-\beta}\ge\frac{\beta\log 2}{2}>\frac{\beta}{3}.$$

It follows that

$$q^2\ge\frac{7c^2\beta K}{24y}\left(\frac{t}{4}\right)^{\beta}. \tag{12}$$

By (7),

$$K\ge\frac{1}{4}\delta^2xy^2,\qquad t=K/y^2\ge\frac{1}{4}\delta^2x.$$

Substituting these inequalities into (12) yields

$$q^2\ge\frac{7}{96}\,16^{-\beta}c^2\beta\,\delta^{4\alpha}x^{2\alpha}y.$$

Since $\beta\le 1$, $16^{-\beta}\ge 1/16$, and

$$\frac{7}{1536}>\frac{1}{256}.$$

Therefore

$$q^2\ge\frac{c^2\beta}{256}\delta^{4\alpha}x^{2\alpha}y,$$

which is exactly (4) after taking square roots. □

5. A fixed-dimensional lower bound

Theorem 5.1. *For every integer $d\ge 1$ and every finite nonempty $F\subset\mathbb{R}^d$,*

$$\boxed{|D(F)|\ge e^{-4d}|F|^{1/2+1/(4d^2-2)}.} \tag{13}$$

Proof. Put

$$\tau_d=\frac{1}{2}+\varepsilon_d,\qquad \varepsilon_d=\frac{1}{4d^2-2}.$$

We induct on d. For $d=1$, if $F=\{x_1<\cdots<x_N\}$, then the N values x_i-x_1 lie in $D(F)$, so $|D(F)|\ge N$. Since $\tau_1=1$, (13) follows.

Let $d\ge 2$, write $N=|F|$, and translate one point to the origin. Assign each of the remaining $N-1$ points to exactly one of the 2^d sign orthants, using a fixed convention for zero coordinates. One orthant, together with the origin, contains a set $B\subset F$ of size

$$m\ge 1+\frac{N-1}{2^d}\ge 2^{-d}N. \tag{14}$$

If all coordinates in this orthant have the same sign, comparison with the origin gives $|D(F)|\ge m$. Since $\tau_d\le 1$ and $2^{-d}\ge e^{-4d}$, the theorem follows in this case.

Otherwise, after replacing the set by its negative if necessary and permuting coordinates, write the points of B as $(u,-v)$, where the positive block has dimension

$$1\le p\le d/2.$$

The replacement $F \mapsto -F$ does not change $|D(F)|$, because $D(-F) = D(F)$. Regard B as the edge set of a rooted graph $G \subset X \times Y$.

By induction, every finite nonempty $W \subset \mathbb{R}^p$ satisfies

$$|D(W)| \geq c_p |W|^{\alpha_p}, \qquad c_p = e^{-4p}, \qquad \alpha_p = \frac{1}{2} + \varepsilon_p.$$

Let $q = |D(F)|$. Lemma 4.1 and Lemma 3.1 give

$$q \geq A_p \delta^{2\alpha_p} x^{\alpha_p} y^{1/2}, \qquad q \geq x, \qquad q \geq y,$$

where

$$A_p = \frac{c_p}{16}\sqrt{2\alpha_p - 1}.$$

Since $m = \delta x y$, multiplying the first inequality by the α_p-th power of $q \geq x$ and the $(2\alpha_p - 1/2)$-th power of $q \geq y$ gives

$$q^{3\alpha_p + 1/2} \geq A_p m^{2\alpha_p}.$$

Thus

$$q \geq C_p m^{T_p}, \qquad C_p = A_p^{1/s_p}, \qquad s_p = 3\alpha_p + \frac{1}{2}, \qquad T_p = \frac{2\alpha_p}{s_p}. \tag{15}$$

A direct calculation gives

$$T_p - \frac{1}{2} = \frac{\varepsilon_p}{4 + 6\varepsilon_p} = \frac{1}{16p^2 - 2}. \tag{16}$$

Since $2p \leq d$, (16) implies $T_p \geq \tau_d$.

By Lemma 2.2, $q \geq m^{1/2}$. Interpolate this with (15). Set

$$\lambda = \frac{\tau_d - 1/2}{T_p - 1/2} = \frac{16p^2 - 2}{4d^2 - 2} \in (0, 1].$$

Then

$$q \geq C_p^{\lambda} m^{\tau_d} \geq \exp(-L_{d,p}) N^{\tau_d},$$

where

$$L_{d,p} = d\tau_d \log 2 + \frac{\lambda}{s_p}\left(4p + \log 16 + \frac{1}{2}\log(2p^2 - 1)\right). \tag{17}$$

Indeed, $2\alpha_p - 1 = (2p^2 - 1)^{-1}$ and (14) accounts for the first term.

Now $s_p \geq 2$, $\lambda \leq 1$, and $p \leq d/2$. Hence the contribution of $4p$ in (17) is at most d. Also

$$\frac{\lambda}{s_p}\left(\log 16 + \frac{1}{2}\log(2p^2 - 1)\right) \leq d \qquad (d \geq 2),$$

which follows, for example, from $\lambda/s_p \leq 1/2$, $p \leq d/2$, and $8\sqrt{2}\, d \leq e^{2d}$ for $d \geq 2$. Finally, $d\tau_d \log 2 \leq d \log 2$. Therefore

$$L_{d,p} \leq (2 + \log 2)d < 4d.$$

This proves (13) and completes the induction. $\square$

6. Small doubling and low affine dimension

We use two standard additive-combinatorial inputs.

Lemma 6.1 (Ruzsa sum-difference inequality)**.** *If A is a nonempty finite subset of an abelian group, then*

$$|A + A| \leq \frac{|A - A|^3}{|A|^2}.$$

This is the standard Ruzsa sum-difference inequality; see, for example, [4, Chapter 2].

Theorem 6.2 (Weak PFR over the integers)**.** *There are absolute constants $B, D_0 > 0$ with the following property. If $A \subset \mathbb{Z}^D$ is finite and nonempty and*

$$|A + A| \le K|A|,$$

then there is a nonempty subset $A' \subset A$ such that

$$|A'| \ge (2K)^{-B}|A| \qquad \text{and} \qquad \dim_{\mathrm{aff}}(A') \le D_0 \log(2K).$$

For $K \ge 2$, this is Theorem 1.3 of Gowers–Green–Manners–Tao [1], which explicitly gives a subset of polynomial relative size and ordinary affine dimension $O(\log K)$. Enlarging the constants and replacing K by $2K$ makes the statement uniform for all $K \ge 1$.

7. Proof of the dimension-free lower bound

Proof of Theorem 1.1. By Proposition 2.1, it is enough to prove the corresponding statement for every finite lattice set. Let $F \subset \mathbb{Z}^D$ be arbitrary with

$$|F| = n.$$

Define $L \ge 1$ and $t \ge 0$ by

$$|D(F)| = L\sqrt{n}, \qquad t = \log L.$$

The inequality $L \ge 1$ follows from Lemma 2.2.

By Lemma 2.2,

$$|F - F| \le |D(F)|^2 = L^2 n.$$

Lemma 6.1 therefore gives

$$|F + F| \le L^6 n.$$

Apply Theorem 6.2 with $K = L^6$. There is a nonempty $F' \subset F$ such that

$$N := |F'| \ge 2^{-B} n L^{-6B}, \qquad d := \dim_{\mathrm{aff}}(F') \le D_0 \log(2L^6). \tag{18}$$

Choose $D_1 = 6D_0$. Since $\log 2 < 1$,

$$d \le D_1(1 + t). \tag{19}$$

If $d = 0$, then $N = 1$, and the first inequality in (18) gives

$$t \ge \frac{\log n - B \log 2}{6B},$$

which is stronger than the desired conclusion for all sufficiently large n. Hence assume $d \ge 1$.

The restrictions of the ambient coordinate functionals span the dual of the direction space of the affine hull of F'. Choose d of them forming a basis. The associated coordinate projection π is injective on F' and, by Lemma 2.2,

$$|D(F)| \ge |D(F')| \ge |D(\pi F')|.$$

Applying Theorem 5.1 to $\pi F' \subset \mathbb{R}^d$ and using

$$\frac{1}{4d^2 - 2} \ge \frac{1}{4d^2},$$

we obtain

$$L\sqrt{n} \ge e^{-4d} N^{1/2 + 1/(4d^2)}. \tag{20}$$

Let $X_0 = \log n$ and put $\theta = 1/(4d^2)$. Using the lower bound for N in (18), taking logarithms in (20), and cancelling $X_0/2$ gives

$$\big(1 + 3B + 6B\theta\big)t + 4d + \big(\tfrac{1}{2} + \theta\big)B \log 2 \ge \theta X_0. \tag{21}$$

Since $d \ge 1$, we have $0 < \theta \le 1/4$. By (19), the left side of (21) is at most

$$C_1(1 + t)$$

for an absolute constant $C_1 > 0$ depending only on B and D_0. On the other hand,

$$\theta X_0 = \frac{X_0}{4d^2} \geq \frac{X_0}{4D_1^2(1+t)^2}.$$

Consequently

$$(1+t)^3 \geq \frac{\log n}{4C_1 D_1^2}.$$

Let

$$c_1 = (4C_1 D_1^2)^{-1/3}.$$

For all sufficiently large n,

$$t \geq \frac{c_1}{2}(\log n)^{1/3}.$$

Since $t = \log(|D(F)|/\sqrt{n})$,

$$|D(F)| \geq \sqrt{n}\,\exp\Big(\frac{c_1}{2}(\log n)^{1/3}\Big).$$

The set F was arbitrary. Minimizing over all F of cardinality n and invoking Proposition 2.1 proves the theorem with $c = c_1/2$. □

8. Dependency and bookkeeping audit

The only external structural input beyond standard additive-combinatorial inequalities is Theorem 6.2. The cited theorem controls ordinary affine dimension, not merely skew-dimension, and its subset loss is polynomial in the doubling constant. In the present application the doubling constant is at most L^6, so the theorem gives exactly the bounds in (18) after renaming absolute constants.

All finite inequalities in Lemma 4.1 use the following conventions. The energy sum in (6) is over ordered pairs; K is the unordered off-diagonal sum; and e_j counts unordered pairs. Hence there are exactly $2e_j$ associated oriented pairs in (9). The dyadic intervals are half-open and partition all pairs with codegree at least $t/4$. Finally, shelves corresponding to distinct labels are disjoint because the complete second coordinate of every shelf element equals its label.

9. Remarks on the result

Remark 9.1 (Why this is a lower bound)*.* No construction is chosen anywhere in the proof of Theorem 1.1. The argument applies to an arbitrary n-point lattice set F, hence to the exponent-vector image of every n-element set of positive integers. Taking the minimum therefore preserves the inequality in the lower-bound direction.

Remark 9.2 (Comparison with the known upper bound)*.* Theorem 1.1 proves

$$\log \frac{h(n)}{\sqrt{n}} \gg (\log n)^{1/3},$$

whereas the upper estimate in (1) gives

$$\log \frac{h(n)}{\sqrt{n}} \ll (\log n)^{1/2}.$$

Thus the result proves in particular that $h(n)/\sqrt{n} \to \infty$ and rules out every fixed polylogarithmic correction, but it does not determine the exact subpolynomial factor.

Remark 9.3 (Verification)*.* The logical proof is entirely contained in the preceding sections. Supplementary code checks exponent identities, constant recurrences, and selected finite graph instances. These computations are not used as proof and do not replace independent review of Lemma 4.1.

References

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